COMP9602: Optimization

Duality (III)

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Next

- Optimality conditions
  - Complementary slackness
  - KKT conditions

Complementary slackness

assume strong duality holds, $x^\star$ is primal optimal, $(\lambda^\star, \nu^\star)$ is dual optimal

$$\begin{aligned} f_0(x^\star) = g(\lambda^\star, \nu^\star) &= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^\star f_i(x) + \sum_{i=1}^p \nu_i^\star h_i(x) \right) \\ &\leq f_0(x^\star) + \sum_{i=1}^m \lambda_i^\star f_i(x^\star) + \sum_{i=1}^p \nu_i^\star h_i(x^\star) \\ &\leq f_0(x^\star) \end{aligned}$$

hence, the two inequalities hold with equality

- $x^\star$ minimizes $L(x, \lambda^\star, \nu^\star)$

- $\lambda_i^\star f_i(x^\star) = 0$ for $i = 1, \ldots, m$ (known as complementary slackness):

$$\lambda_i^\star > 0 \Longrightarrow f_i(x^\star) = 0, \qquad f_i(x^\star) < 0 \Longrightarrow \lambda_i^\star = 0$$

Complementary slackness for LP

- Primal and dual LPs $(x \in R^n, y \in R^m, A : m \times n)$

Primal: $\min c^T x$

subject to:

$$Ax \succeq b$$

$$x \succeq 0$$

Canonical form LP

Dual: $\max b^T y$

subject to:

$$A^T y \preceq c$$

$$y \succeq 0$$

- Complementary slackness theorem for LP

A pair of feasible solutions $x \in R^n$ and $y \in R^m$ for primal and dual LP problems is optimal if and only if

$$y_i(b_i - (Ax)_i) = 0, \forall i = 1, \ldots, m$$

$$x_j(c_j - (A^T y)_j) = 0, \forall j = 1, \ldots, n$$

$$\Updownarrow$$

$$(Ax)_i = b_i, \text{ or } (Ax)_i > b_i \ \& \ y_i = 0, \forall i = 1, \ldots, m$$

$$(A^T y)_j = c_j, \text{ or } (A^T y)_j < c_j \ \& \ x_j = 0, \forall j = 1, \ldots, n$$

Complementary slackness for LP

- Proof of sufficiency:

Given x, y satisfy complementary slackness

$$y_i(b_i - (Ax)_i) = 0, \forall i \iff b^T y = y^T(Ax)$$

$$x_j(c_j - (A^T y)_j) = 0, \forall j \iff c^T x = x^T A^T y = y^T Ax$$

Hence $c^T x = b^T y$

We have $c^T x = b^T y \leq b^T y^* \leq c^T x^* \leq c^T x$

Therefore, x is primal optimal and y is dual optimal

Complementary slackness for LP

- Proof of necessity:

  Given x is primal optimal and y is dual optimal, from general complementary slackness of strong duality

$$y_i(b_i - (Ax)_i) = 0, \forall i = 1, \ldots, m$$

$$\lambda_j x_j = 0, \forall j = 1, \ldots, n$$

=> $x_j(c_j - (A^T y)_j) = 0, \forall j = 1, \ldots, n$

Karush-Kuhn-Tucker (KKT) conditions

the following four conditions are called KKT conditions (for a problem with differentiable f_i, h_i):

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

1. primal constraints: $f_i(x) \leq 0$, $i = 1, \ldots, m$, $h_i(x) = 0$, $i = 1, \ldots, p$
2. dual constraints: $\lambda \succeq 0$
3. complementary slackness: $\lambda_i f_i(x) = 0$, $i = 1, \ldots, m$
4. gradient of Lagrangian with respect to x vanishes:

$$\nabla f_0(x) + \sum_{i=1}^{m} \lambda_i \nabla f_i(x) + \sum_{i=1}^{p} \nu_i \nabla h_i(x) = 0$$

Karush-Kuhn-Tucker (KKT) conditions

- For any optimization problem with differentiable objective and constraint functions, if strong duality holds, any pair of primal and dual optimal points x^*, λ^*, ν^* must satisfy the KKT conditions

$$g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) = \inf_x f_0(x) + \sum_{i=1}^{m} \lambda_i^* f_i(x) + \sum_{i=1}^{p} \nu_i^* h_i(x) = f_0(x^*)$$

=> x^* minimizes $L(x, \lambda^*, \nu^*)$

KKT conditions for convex programs

- For a convex program, any points $\tilde{x}, \tilde{\lambda}, \tilde{\nu}$ that satisfy the KKT conditions are primal and dual optimal, and have zero duality gap

$\tilde{\lambda}_i \geq 0, \forall i$ => $L(x, \tilde{\lambda}, \tilde{\nu})$ is convex in x

gradient of Lagrangian with respect to $\tilde{x}$ vanishes => $\tilde{x}$ minimizes $L(x, \tilde{\lambda}, \tilde{\nu})$

$$\text{then } g(\tilde{\lambda}, \tilde{\nu}) = \inf_x L(x, \tilde{\lambda}, \tilde{\nu}) = f_0(\tilde{x}) + \sum_{i=1}^{m} \tilde{\lambda}_i f_i(\tilde{x}) + \sum_{i=1}^{p} \tilde{\nu}_i h_i(\tilde{x})$$

$$= f_0(\tilde{x})$$

since $g(\tilde{\lambda}, \tilde{\nu}) \leq g(\lambda^*, \nu^*) \leq f_0(x^*) \leq f_0(\tilde{x})$

we know $\tilde{x}, \tilde{\lambda}, \tilde{\nu}$ are primal and dual optimal, respectively

KKT conditions for convex programs (cont'd)

- If a convex program with differentiable objective and constraint functions satisfies Slater's conditions, then KKT condition is necessary and sufficient for optimality

  - recall that Slater implies strong duality

Importance of KKT conditions

- KKT conditions are very important in optimization
  - possibly solve the KKT conditions to derive (analytical) solution for primal/dual problem
  - many algorithms for convex optimization can be interpreted as methods for solving the KKT conditions

A simple example

$$\min x_1^2 + 2x_2^2$$

subject to:

$$x_1 + x_2 \geq 3$$

To solve this, try to find (x^*, λ^*) that satisfy the KKT conditions

Water-filling examples

$$\min -\sum_{i=1}^{k} \log(1 + \frac{p_i}{N_i})$$

subject to:

$$\sum_{i=1}^{k} p_i \leq P$$

$$p_i \geq 0, \forall i = 1, \ldots, k$$

Another: Example 5.2, pp. 245, textbook

Water-filling examples (cont'd)

KKT conditions:

$$\sum_{i=1}^{k} p_i \leq P$$

$$p_i \geq 0, \forall i = 1, \ldots, k$$

$$\lambda \geq 0 \quad \mu_i \geq 0, \forall i = 1, \ldots, k$$

$$\lambda(\sum_{i=1}^{k} p_i - P) = 0$$

$$\mu_i p_i = 0, \forall i = 1, \ldots, k$$

$$\frac{\partial L(p, \lambda, \mu)}{\partial p_i} = 0 \quad \Rightarrow \quad \frac{1}{\lambda - \mu_i} = p_i + N_i, \forall i = 1, \ldots, k$$

$$L(p, \lambda, \mu) = -\sum_{i=1}^{K} \log(1 + \frac{p_i}{N_i}) + \lambda(\sum_{i=1}^{k} p_i - P) + \sum_{i=1}^{k} \mu_i(-p_i)$$

Water-filling examples (cont'd)

According to KKT conditions:

If $p_i > 0,\ \mu_i = 0\ ,\quad p_i = \dfrac{1}{\lambda - \mu_i} - N_i = \dfrac{1}{\lambda} - N_i$

If $p_i = 0,\ p_i + N_i = \dfrac{1}{\lambda - \mu_i} \geq \dfrac{1}{\lambda}$ => $\lambda \geq \dfrac{1}{N_i}$

Hence $p_i = \begin{cases} \frac{1}{\lambda} - N_i, \lambda < \frac{1}{N_i} \\ 0, \lambda \geq \frac{1}{N_i} \end{cases}$

How to decide λ: make use of $\sum_{i=1}^{k} p_i = P$

Water-filling examples: generalization

$$\min f(p)$$

subject to:

$$\sum_{i=1}^{k} p_i \leq P$$

$$p_i \geq 0, \forall i = 1, \ldots, k$$

where f(p) is convex and twice differentiable ($\bigtriangledown^2 f(p) \succ 0$), $\frac{\partial f(\mathbf{p})}{\partial p_i} < 0$ and invertible, $f(p)$ is separable in terms of each p_i ,

the same water-filling algorithm applies:

maximally allocate resource to pi with the current smallest marginal utility ($\frac{\partial f(\mathbf{p})}{\partial p_i}$,that increases with pi), until all resource P is used up.

Water-filling examples: generalization (cont'd)

$$L(p,\lambda,\mu) = f(p) + \lambda(\sum_{i=1}^{k} p_i - P) + \sum_{i=1}^{k} \mu_i(-p_i)$$

$$\frac{\partial L(p,\lambda,\nu)}{\partial p_i} = \frac{\partial f(p)}{\partial p_i} + \lambda - \mu_i = 0$$

If $p_i > 0,\ \mu_i = 0,\ \dfrac{\partial f(p)}{\partial p_i} + \lambda = 0$

If $p_i = 0,\ \left.\dfrac{\partial f(p)}{\partial p_i}\right|_0 + \lambda \geq 0$

let $f_i'(p) = \dfrac{\partial f(p)}{\partial p_i}$

Hence $p_i = \begin{cases} f_i'^{-1}(-\lambda), \lambda < -f_i'(0) \\ 0, \lambda \geq -f_i'(0) \end{cases}$

- Reference
  - Chapter 5.5, Convex Optimization.
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